

GWALIOR, India

WMO: 423610

Lat: 26.2112N

Lon: 78.2024E

Elev: 188

StdP: 99.09

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	7.0	0.2	3.9	22.8	2.1	4.5	19.5	4.6	18.4	3.5	18.3	0.1	140	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.8	43.8	23.4	42.6	23.4	41.3	23.3	28.7	33.6	28.2	33.0	27.8	32.5	2.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	24.1	30.9	27.2	23.5	30.4	26.8	23.0	30.1	95.0	33.8	92.9	33.2	91.0	32.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.4	3.4	4.0	45.5	1.5	1.0	2.9	46.2	2.1	46.8	1.2	47.3	0.2	48.0		
			3.3	29.8	1.3	0.6	2.3	30.2	1.5	30.5	0.8	30.9	-0.2	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	15.0	19.2	25.1	30.8	34.5	34.5	30.8	29.5	29.3	26.7	21.7	16.9
	DBStd	6.81	2.66	2.82	3.08	2.75	2.32	3.10	2.34	1.55	1.59	2.18	2.22	2.57
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	191	109	21	1	0	0	0	0	0	0	0	2	58
	CDD10.0	5908	156	256	467	624	758	734	644	604	579	519	352	214
	CDD18.3	3056	5	44	209	374	500	484	386	346	329	261	104	14
	CDH23.3	41801	72	483	2409	5467	8294	7855	5260	4310	3977	2638	886	150
Wind	WSAvg	24591	8	144	1303	3563	5884	5477	2818	1912	1834	1326	305	17
	WSAvg	0.8	0.4	0.8	0.8	1.0	1.3	1.4	1.3	1.1	0.9	0.4	0.2	0.3
	PrecAvg	794	11	14	10	5	12	78	235	228	162	21	6	13
	PrecMax	1096	58	106	123	29	45	248	386	409	539	129	82	139
	PrecMin	507	0	0	0	0	0	3	22	56	22	0	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	PrecStd	192	14	24	23	7	11	66	86	94	113	34	17	32
	DB	27.7	32.9	39.0	43.2	45.0	45.6	40.9	36.9	36.9	36.7	33.2	28.6	
	MCWB	17.5	19.0	21.4	21.7	23.6	22.9	25.7	26.1	24.8	22.6	19.8	17.9	
	DB	25.3	30.4	36.9	41.8	43.9	43.9	38.7	34.9	35.7	35.4	31.3	26.7	
	MCWB	16.3	18.8	19.8	21.2	23.2	24.2	26.3	26.9	25.5	22.7	19.5	16.8	
	DB	23.3	28.3	35.2	40.4	42.9	42.5	36.6	33.8	34.7	34.3	30.0	25.0	
	MCWB	15.0	17.6	19.9	21.0	23.1	24.3	26.6	27.2	25.8	22.6	19.3	16.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	DB	21.4	26.5	33.4	39.1	41.8	40.9	35.0	32.7	33.6	33.0	28.6	23.2	
	MCWB	14.2	16.8	19.6	21.3	23.1	24.8	26.8	27.1	25.8	22.6	18.8	15.7	
	WB	19.2	21.4	24.4	26.1	28.4	29.0	29.2	29.1	28.9	27.2	22.8	19.7	
	MCDB	25.3	28.8	35.1	37.8	40.4	36.5	33.6	32.7	33.2	32.5	28.6	25.3	
	WB	17.5	19.9	22.4	24.2	26.9	28.2	28.6	28.4	28.1	26.2	21.8	18.3	
	MCDB	23.3	27.6	33.0	37.2	38.7	35.7	33.0	32.1	32.4	31.0	28.0	24.6	
	WB	16.1	18.8	21.2	23.0	25.8	27.6	28.2	28.0	27.6	25.2	20.7	17.3	
Mean Daily Temperature Range	MCDB	21.2	26.3	31.5	36.4	37.6	34.9	32.5	31.7	31.6	30.0	26.9	23.4	
	WB	15.0	17.8	20.2	22.1	24.9	27.1	27.8	27.6	27.0	24.4	19.7	16.3	
	MCDB	19.5	24.5	30.3	35.1	37.2	34.1	31.9	31.0	30.9	29.5	26.6	21.8	
	MDBR	12.7	14.0	15.1	15.4	13.8	10.2	6.6	5.9	7.9	13.2	15.1	14.0	
	MCDBR	15.7	16.1	17.1	16.5	14.8	12.3	8.9	7.5	10.1	14.3	15.9	15.9	
Clear-Sky Solar Irradiance	MCWBR	7.4	6.5	5.6	5.3	4.6	3.5	2.6	2.8	3.4	5.1	6.5	7.2	
	MCDBR	13.2	13.9	15.4	15.5	13.5	9.7	6.9	6.4	7.4	10.6	13.8	14.4	
	MCWBR	7.0	6.5	6.0	6.2	5.3	3.4	2.9	2.9	3.3	4.7	6.4	7.2	
	taub	0.498	0.475	0.490	0.560	0.636	0.701	0.693	0.619	0.563	0.599	0.591	0.555	
All-Sky Solar Radiation	taud	1.860	1.971	1.966	1.830	1.715	1.637	1.670	1.844	1.942	1.761	1.707	1.748	
	Ebn at Noon	727	794	809	762	704	656	660	709	738	673	635	648	
	Edh at Noon	179	172	182	214	240	257	249	208	184	209	207	193	
All-Sky Solar Radiation	RadAvg	3.53	4.75	6.02	6.82	6.92	5.92	4.62	4.63	5.08	4.92	3.95	3.45	
	RadStd	0.39	0.31	0.23	0.21	0.26	0.45	0.43	0.38	0.46	0.28	0.29	0.19	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page